

VIETNAM NATIONAL UNIVERSITY HO CHI MINH CITY

HO CHI MINH CITY UNIVERSITY OF TECHNOLOGY

Faculty of Electrical and Electronics Engineering



Microwave Engineering and Systems

EE5430

Assignment 1

Transmission-Line Parameters and Wave Propagation

Student: Nguyen Thai Thanh Binh

Student ID: 2570175

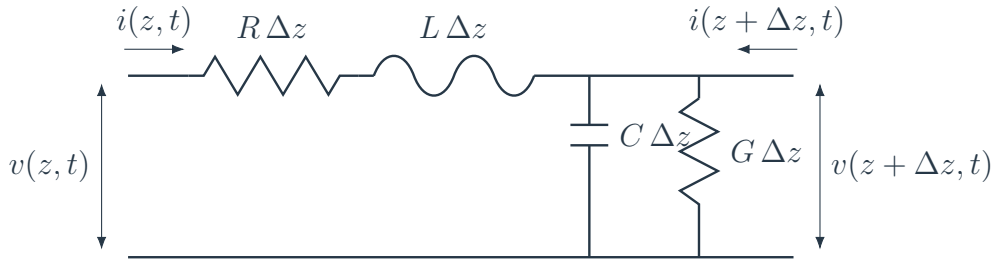
Report version: Analytical derivation

September 2026

Contents

1	Problem 1: Parameters of a lossy transmission line	2
2	Problem 2: A nominal 50-ohm coaxial cable at 500 MHz	2
2.1	Exact calculation	3
2.2	Low-loss approximation and comparison	3
2.3	Phase velocity and wavelength	3
3	Problem 3: Matched, lossless line	4
4	Problem 4: Matched air line and an open-circuit comparison	5
4.1	Instantaneous voltage and current	5
4.2	Average load power	5
4.3	If the load is replaced by an open circuit	5
	Final numerical summary	6

1 Problem 1: Parameters of a lossy transmission line



The distributed parameters are

$$R = 1.5 \, \Omega/\text{m}, \quad G = 0.4 \, \text{mS}/\text{m}, \quad L = 10 \, \text{nH}/\text{m}, \quad C = 0.2 \, \text{pF}/\text{m}, \quad f = 2 \, \text{GHz}.$$

With the $e^{j\omega t}$ convention, the exact characteristic impedance and propagation constant are

$$Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}}, \quad (1.1)$$

$$\gamma = \alpha + j\beta = \sqrt{(R + j\omega L)(G + j\omega C)}. \quad (1.2)$$

At $\omega = 2\pi f = 1.2566371 \times 10^{10} \, \text{rad/s}$,

$$R + j\omega L = 1.5 + j125.6637 \, \Omega/\text{m},$$

$$G + j\omega C = 4.0000 \times 10^{-4} + j2.51327 \times 10^{-3} \, \text{S}/\text{m}.$$

Therefore,

$$Z_0 = 221.6297 + j16.1961 \, \Omega = 222.2207 \angle 4.1796^\circ \, \Omega, \quad (1.3)$$

$$\gamma = 0.0479467 + j0.5634947 \, \text{m}^{-1}. \quad (1.4)$$

Consequently,

$$\alpha = 0.0479467 \, \text{Np}/\text{m},$$

$$\alpha_{\text{dB}} = 8.6858896\alpha = 0.41646 \, \text{dB}/\text{m},$$

$$\beta = 0.563495 \, \text{rad}/\text{m},$$

$$\lambda = \frac{2\pi}{\beta} = 11.1504 \, \text{m}.$$

Answer. $Z_0 = 221.63 + j16.20 \, \Omega$ and $\gamma = (0.04795 + j0.56349) \, \text{m}^{-1}$. Thus the line attenuates at $0.04795 \, \text{Np}/\text{m} = 0.4165 \, \text{dB}/\text{m}$, while $\beta = 0.56349 \, \text{rad}/\text{m}$ and $\lambda = 11.150 \, \text{m}$.

2 Problem 2: A nominal 50-ohm coaxial cable at 500 MHz

Here

$$R = 4 \, \Omega/\text{m}, \quad G = 1 \, \text{mS}/\text{m}, \quad L = 250 \, \text{nH}/\text{m}, \quad C = 100 \, \text{pF}/\text{m}, \quad f = 500 \, \text{MHz}.$$

2.1 Exact calculation

For $\omega = 3.1415927 \times 10^9$ rad/s,

$$R + j\omega L = 4 + j785.3982 \, \Omega/\text{m},$$

$$G + j\omega C = 0.001 + j0.3141593 \, \text{S}/\text{m}.$$

Substitution in the exact formulas gives

$$Z_0 = 50.0001748 - j0.0477458 \, \Omega = 50.0001976 \angle (-0.054712^\circ) \, \Omega, \quad (2.1)$$

$$\gamma = 0.06499997 + j15.70797043 \, \text{m}^{-1}. \quad (2.2)$$

Thus $\alpha = 0.06499997 \, \text{Np}/\text{m} = 0.5645897 \, \text{dB}/\text{m}$ and $\beta = 15.70797043 \, \text{rad}/\text{m}$.

2.2 Low-loss approximation and comparison

The loss ratios are

$$\frac{R}{\omega L} = 5.093 \times 10^{-3} \ll 1, \quad \frac{G}{\omega C} = 3.183 \times 10^{-3} \ll 1,$$

so the low-loss formulas apply:

$$Z_0 \simeq \sqrt{\frac{L}{C}} = 50.000 \, \Omega, \quad (2.3)$$

$$\alpha \simeq \frac{1}{2} \left(\frac{R}{Z_0} + GZ_0 \right) = \frac{1}{2}(0.08 + 0.05) = 0.065000 \, \text{Np}/\text{m}, \quad (2.4)$$

$$\beta \simeq \omega \sqrt{LC} = 15.7079633 \, \text{rad}/\text{m}. \quad (2.5)$$

Quantity	Exact	Low-loss	Relative difference
$ Z_0 \, (\Omega)$	50.0001976	50.0000000	$3.95 \times 10^{-4}\%$
$\alpha \, (\text{Np}/\text{m})$	0.06499997	0.06500000	$4.56 \times 10^{-5}\%$
$\beta \, (\text{rad}/\text{m})$	15.70797043	15.70796327	$4.56 \times 10^{-5}\%$

2.3 Phase velocity and wavelength

Using the exact phase constant,

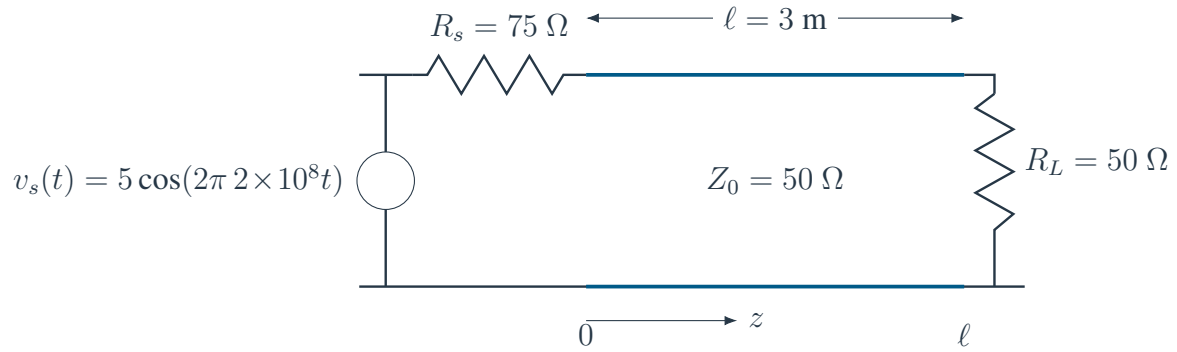
$$v_p = \frac{\omega}{\beta} = 1.9999991 \times 10^8 \, \text{m}/\text{s}, \quad (2.6)$$

$$\lambda = \frac{2\pi}{\beta} = 0.3999998 \, \text{m}. \quad (2.7)$$

The low-loss values are $v_p \simeq 1/\sqrt{LC} = 2.0000 \times 10^8 \, \text{m}/\text{s}$ and $\lambda \simeq 0.4000 \, \text{m}$.

Answer. Exact: $Z_0 = 50.00017 - j0.04775 \, \Omega$ and $\gamma = (0.06499997 + j15.70797043) \, \text{m}^{-1}$. The low-loss approximation is essentially identical; $v_p \approx 2.00 \times 10^8 \, \text{m}/\text{s}$ and $\lambda \approx 0.400 \, \text{m}$.

3 Problem 3: Matched, lossless line



The load is matched, so $\Gamma_L = 0$ and only a forward wave exists. The source frequency and the line constants are

$$\begin{aligned} f &= 2.0 \times 10^8 \text{ Hz}, & \omega &= 4\pi \times 10^8 \text{ rad/s}, \\ v_p &= 2.0 \times 10^8 \text{ m/s}, & \lambda &= \frac{v_p}{f} = 1.0 \text{ m}, \\ \beta &= \frac{2\pi}{\lambda} = 2\pi \text{ rad/m}. \end{aligned}$$

Because the input impedance of a matched line equals Z_0 , the incident-wave amplitude follows from a voltage divider:

$$V_0^+ = 5 \frac{Z_0}{R_s + Z_0} = 5 \frac{50}{75 + 50} = 2.0 \text{ V}, \quad I_0^+ = \frac{V_0^+}{Z_0} = 0.040 \text{ A}.$$

Taking $z = 0$ at the line input, the instantaneous waves are therefore

$$v(z, t) = 2 \cos(4\pi \times 10^8 t - 2\pi z) \text{ V}, \quad (3.1)$$

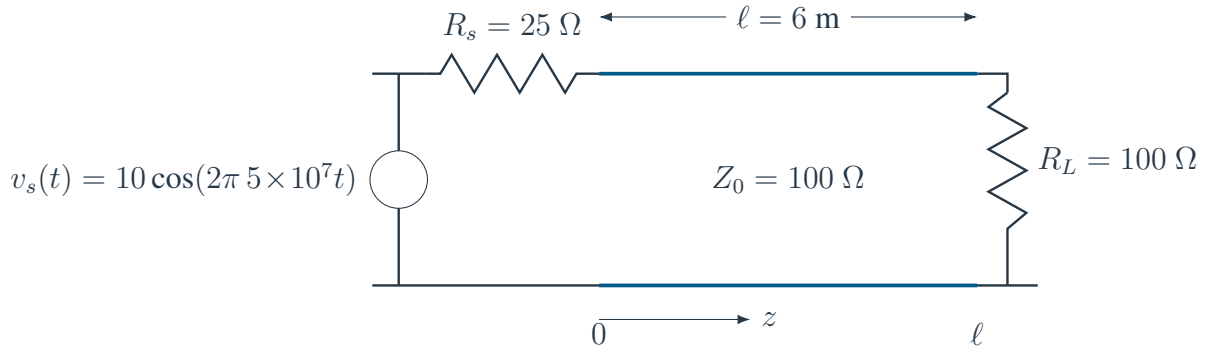
$$i(z, t) = 0.040 \cos(4\pi \times 10^8 t - 2\pi z) \text{ A}, \quad 0 \leq z \leq 3 \text{ m}. \quad (3.2)$$

Since $\ell/\lambda = 3$, the electrical length is $\beta\ell = 6\pi \text{ rad} = 1080^\circ$. At the load,

$$v_L(t) = v(3, t) = 2 \cos(4\pi \times 10^8 t - 6\pi) = 2 \cos(4\pi \times 10^8 t) \text{ V}.$$

Answer. The line is three wavelengths long. Its load voltage has a peak value of 2 V and is in phase with the source voltage (the accumulated phase is 1080° , an integer number of cycles).

4 Problem 4: Matched air line and an open-circuit comparison



For an air line, $v_p = 3.0 \times 10^8$ m/s. Hence

$$f = 5.0 \times 10^7 \text{ Hz},$$

$$\omega = \pi \times 10^8 \text{ rad/s},$$

$$\lambda = \frac{v_p}{f} = 6.0 \text{ m},$$

$$\beta = \frac{2\pi}{\lambda} = \frac{\pi}{3} \text{ rad/m}.$$

4.1 Instantaneous voltage and current

For the matched load, $Z_{\text{in}} = Z_0 = 100 \Omega$ and

$$V_0^+ = 10 \frac{100}{25 + 100} = 8.0 \text{ V}, \quad I_0^+ = \frac{8}{100} = 0.080 \text{ A}.$$

Thus

$$v(z, t) = 8 \cos \left(\pi \times 10^8 t - \frac{\pi}{3} z \right) \text{ V}, \quad (4.1)$$

$$i(z, t) = 0.080 \cos \left(\pi \times 10^8 t - \frac{\pi}{3} z \right) \text{ A}, \quad 0 \leq z \leq 6 \text{ m}. \quad (4.2)$$

4.2 Average load power

The load voltage has peak amplitude 8 V, so

$$P_{L, \text{av}} = \frac{V_{L, \text{rms}}^2}{R_L} = \frac{(8/\sqrt{2})^2}{100} = 0.320 \text{ W}. \quad (4.3)$$

4.3 If the load is replaced by an open circuit

The forward-wave expression above is no longer valid because an open circuit has $\Gamma_L = +1$. A reflected wave of equal voltage amplitude is produced. Here $\ell = \lambda$, so $Z_{\text{in}} \rightarrow \infty$ and the source current is zero; therefore the total input voltage equals the source voltage. With $\tilde{V}_0^- = \tilde{V}_0^+ = 5 \text{ V}$,

$$\tilde{V}(z) = 5e^{-j\beta z} + 5e^{j\beta z} = 10 \cos(\beta z), \quad (4.4)$$

$$v(z, t) = 10 \cos \left(\frac{\pi z}{3} \right) \cos(\pi \times 10^8 t) \text{ V}. \quad (4.5)$$

This is a standing wave, not a single traveling wave.

Answer. For the matched load, $v(z, t) = 8 \cos(\pi 10^8 t - \pi z/3)$ V, $i(z, t) = 0.08 \cos(\pi 10^8 t - \pi z/3)$ A, and $P_L = 0.320$ W. With an open circuit the formula changes to a standing-wave expression because $\Gamma_L = +1$.

Final numerical summary

Problem	Main result
1	$Z_0 = 221.63 + j16.20 \, \Omega$; $\gamma = 0.04795 + j0.56349 \, \text{m}^{-1}$; $\lambda = 11.150 \, \text{m}$.
2	$Z_0 = 50.00017 - j0.04775 \, \Omega$; $\gamma = 0.06500 + j15.70797 \, \text{m}^{-1}$; $v_p \approx 2.00 \times 10^8 \, \text{m/s}$, $\lambda \approx 0.400 \, \text{m}$.
3	$v = 2 \cos(4\pi 10^8 t - 2\pi z)$ V, $i = 0.04 \cos(4\pi 10^8 t - 2\pi z)$ A; $\ell = 3\lambda$.
4	$v = 8 \cos(\pi 10^8 t - \pi z/3)$ V, $i = 0.08 \cos(\pi 10^8 t - \pi z/3)$ A; $P_L = 0.320$ W.